our Special Issue are all Manifestations. Finally, the Item is an actual copy of the Manifestation.

Next up, the rights—there are four aspects to consider. Firstly, the Permissions—what you're allowed to do with the content. For example, you're allowed to read this article, but not copy any content from it. Closely tied to Permissions are Constraints, which put limits on your Permissions. In similar vein—you may use some content from the article (permission), but you *must* quote the source of this content (constraint). Then there are Obligations, which you must fulfil before getting access to your Permissions. You've paid for this magazine, so your most important Obligation is out of the way. Finally, all these are mapped to a Rights Holder. Different persons have different rights over media—while you don't have the permission to reproduce this article, Jasubhai Digital Media is free to do so.

Coming Together

All right, so we've made the pretty diagrams—time to turn it into reality. Note that all the information we talked about in the models is *metadata*—it isn't actual content, just information to supplement it. The job of the DRM implementation (effectively just code embedded into the file) has the following things to do:

1. Verify that all the Obligations have been fulfilled—that the content has been paid for and being used by the buyer,
2. Decrypt the content for use, and
3. Track the use and see that it conforms to the Permissions.

In real terms, metadata is expressed inside a special XML document, which conforms to a specific structure called a *Rights Expression Language (REL)*. The structure of XML permits the easy realisation of the models we discussed earlier; moreover, its plain-text nature makes it light and compressible to miniscule degrees.

One REL that hopes to become a universal open standard is the W3C's Open Digital Rights Language (ODRL)—read about it at **http://odrl.net**. An open DRM standard will enable protected content to run on any system without problems—something we don't see in the crowd of DRM technologies we have to deal with today.

Our Top 5 DRM Boo-boo List

5 RealNetworks' Harmony vs. Apple: They made their Harmony DRM system work with the iPod; Apple disabled it in the next firmware update. They made it work again, and Apple disabled it again. Then they gave up.

4 4. Owner Exclusive e-Books in Microsoft Reader: They even put restrictions on text-to-speech!

3 3. StarForce deemed malware: It installed without warning, didn't uninstall, and took a toll on IDE performance.

2 2. The Sony rootkit incident: Shame if you haven't heard of this before, do a quick Google or visit **http://snipurl.com/17uk1**

1 1. The Zune dumbness: Music bought from the Zune marketplace doesn't work on PlaysForSure devices, and PlaysForSure doesn't play on the Zune, despite both technologies being from Microsoft! Talk about shooting yourself in the foot.

Cryptography

Suppose you wanted to send the number 50 to your friend, but want to ensure that nobody who intercepts the message on the way knows what number you're sending. You could multiply it by 6, add 4, divide by 3 and subtract 9. Anyone who sees the message now will think you sent 92.33333. The process is called *encryption*, and performs a whole host of mathematical operations on data to make it unrecognisable. In *decryption*, these mathematical operations are just reversed to recover the original data. Of course, the system fails if anyone figures out the encryption algorithm. The next secure way for encryption is the use of a "key"—a number that is used for these mathematical operations. Without the key, decryption isn't possible, so as long as nobody has the key, your message remains secure.

The most popular method for secure encryption is the Public Key-Private Key system. Think of it as a box with two keys—anyone with the public key can lock it, but only the person with the private key can unlock it. When you register yourself at an online music store, for example, it gives your media player a public and private key—unique to you. When you request a song, your media player sends over the public key, which is then used to encrypt the song you requested. Only your private key can decrypt this song, so unauthorised media players (ones without your private key) can't decrypt it.

Fair Use And Other Stories

To understand what the DRM fuss is all about, you need to understand one aspect of copyright law in particular—fair use. It's one of those vague, infinitely exploitable loopholes that every Copyright Act is plagued with.

To dumb it down considerably, every restriction specified in the act is forgiven, as long as it constitutes a "fair use" of the content. In the early days of the videocassette, courts in the US ruled that taping a TV programme for later viewing constituted fair use—the consumer *has* paid his subscription amount and is thus entitled to the content. Copyright law, on the other hand, expressly states that the content may not be reproduced at all. What makes fair use even more of a mess is that every country has a different definition of it. Ripping music off a CD has widely been accepted as fair use, but giving that ripped music to your friend is a no-no. Letting your friend *borrow* the original CD is fine, however. Go figure! And yet, some countries won't even accept the act of ripping as legal.

Still, fair use clauses are essential—imagine being charged a "performance royalty" for whistling your favourite tune in public. Even more complications arise because the definition of fair use is constantly evolving: in 1980, for example, nobody would have imagined that you could record live TV to your PC, so the definition had to be expanded to include that.

If you're worried about whether your usage counts as "fair", fear not. Here are questions you need to ask yourself:

1. What is the purpose of this use? Personal use is often counted as fair, but the second you enter the realm of distribution and commercial gain, you're breaching copyright law. For example, you can use a song from your music collection to make a remix, but you can't sell it.

2. What is the nature of the work you're using? Creative work—movies, music, articles,